Technical Memo: Direct Vocabulary Steering in Larger Dual-Stream Models

GPT-OSS Interpretability Project

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Abstract

This memo summarizes a first larger-scale evaluation of direct vocabulary steering in locally trained dual-stream and CASCADE-family language models. The intervention is simple: for a binary contrast with first divergent answer tokens token_A and token_B, inject the exact vocabulary direction

$$\Delta v = W[\text{token}_A] - W[\text{token}_B]$$

into the contextual stream and measure the resulting change in local answer preference. We evaluate five GPT-2-vocabulary models ranging from a matched 71M-parameter single-stream/CASCADE pair to larger 12-layer CASCADE+PLS variants. The core result is positive: the effect is sign-consistent, local under the current decomposition, and reasonably disciplined with respect to off-target drift across all evaluated models. The strongest scientific comparison is the matched 71M pair, where CASCADE is slightly stronger than the single-stream baseline on two of three tasks, but not by a large enough margin to justify a strong architecture-superiority claim. The current bottleneck is not runtime but panel quality: one capitalization case remains tokenizer-dirty and should be replaced before this result is treated as a clean steering benchmark.

1 Memo Context

This document is a technical memo in the style of a compact steering paper. It is not intended as a full conference submission. The aim is to record the question, design, current evidence, and next experimental decisions in a form that is easy to cite internally and easy to extend into a later paper.

Implementation status. The repo now has a scaffolded package split between `gpt_oss_interp/steering/`, `gpt_oss_interp/distillation/`, and `gpt_oss_interp/common/`. This matters for the next phase of work: per-channel probing, readout decomposition, and symbolic-slice interventions should now be implemented natively under `steering/`, while teacher-student and CASCADE-target code should live under `distillation/`. The split is still a scaffold, not a completed code migration, but it establishes the intended structure for future work.

Model aliases. To keep the tables compact, we use short aliases:

- **SS-71**: matched 71M single-stream GPT-2-vocab model
- **C-71**: matched 71M CASCADE GPT-2-vocab model
- **CP-71**: 71M CASCADE+PLS GPT-2-vocab model

- **CP-80**: larger CASCADE+PLS Codelion-80M model
- **CP-124**: larger CASCADE+PLS GPT-2-base-scale model

2 Introduction

Standard activation steering work usually applies directions in residual space or hidden-state space, where token identity, context, syntax, and latent task variables are entangled. The motivating idea behind the present line of work is that dual-stream or symbolic-transformer-style architectures may support a more direct and more interpretable intervention. If vocabulary geometry remains legible inside the model, then a human-readable token contrast should function as a usable steering direction rather than merely as a diagnostic object.

The concrete claim examined here is narrower than a general claim about model control. We ask whether the exact vocabulary contrast between two answer tokens can be injected into the contextual stream and produce a predictable local change in answer preference. We also ask whether this remains true in larger models, and whether CASCADE appears cleaner than a matched single-stream baseline.

3 Related Framing

This memo sits at the intersection of three lines of prior internal work.

Causal specialization and controllability. Earlier capitalization-ablation work argued that per-layer supervision (PLS) creates heads with larger and more localized causal effects, summarized under the "hydra effect" framing: standard models route around local damage, whereas more modular models do not. The current steering experiment tests a closely related but distinct question: whether a clean intervention direction exists, not merely whether ablation damage is localized.

Bregman geometry and steerability. The Bregman-geometry notebooks proposed that PLS and CASCADE may improve the conditioning of the output geometry and therefore make steering easier or more disciplined. The present experiment does not directly estimate Hessian conditioning. Instead, it asks whether the resulting intervention behavior already looks local and low-crosstalk in practice.

Architecture-specific interpretability. The broader thesis behind symbolic and dual-stream models is that if a token space remains structurally meaningful inside the model, then direct vocabulary-based interventions should become more plausible than they are in a standard residual-stream architecture.

4 Experimental Question

We evaluate the following question:

> Can a larger dual-stream or CASCADE-family model be steered by injecting an exact vocabulary-space contrast into the contextual stream, and does this yield sign-consistent, local, and reasonably selective behavioral change?

This decomposes into four operational tests:

1. **Sign correctness:** positive and negative scales move the local answer preference in opposite predicted directions.
2. **Locality:** the effect appears at the first divergent token rather than being mediated primarily by suffix or tail behavior.
3. **Collateral discipline:** off-target effects remain smaller than the on-target effect.
4. **Architecture sensitivity:** a matched CASCADE model may show a cleaner response than a matched single-stream baseline.

5 Methods

5.1 Intervention

For a case with binary choices A and B, let token_A and token_B denote the first divergent completion tokens under the active tokenizer. Let W denote the model's tied embedding/unembedding matrix. The steering direction is

$$\Delta v = W[\text{token}_A] - W[\text{token}_B].$$

For scale α, we inject $\alpha \Delta v$ into the contextual stream x_e after a selected transformer block. This is a direct vocabulary intervention rather than a learned steering vector.

Scope note. All experiments in this memo use Δv as one whole d_{model} vector. We do *not* yet exploit a stronger dual-stream-specific intervention family in which the symbolic stream x_t is channelized by head and token embeddings can be written per head or composed across heads. That intervention family is unavailable in a standard transformer and remains an important next step. The correct way to open that space is not by arbitrary head mixing. It requires ACL-style differential probing first: identify which channels behave like entity trackers, recency channels, answer selectors, or generic amplifiers, then intervene on one channel at a time before attempting mixed-token channel composites.

Implementation note. The intended home for that next-phase work is now `gpt_oss_interp/steering/`: probing in `steering/probing.py`, causal symbolic-slice interventions in `steering/interventions.py`, null baselines in `steering/controls.py`, and readout decomposition in `steering/readouts.py`. Shared artifact schemas should remain in `gpt_oss_interp/common/artifacts.py`.

5.2 Evaluation Panel

The same three-case panel was used in all runs:

- `caps_005`
- `induction_009`
- `coref_010`

All large-model runs used raw GPT-2 tokenization. This fixed the pathological reduced-vocabulary collapse seen in the earlier small-model run, especially for `coref_010`. One capitalization issue remains: `caps_005` still diverges at `" d"` vs. `" Dakota"` rather than at a clean full-token capitalization contrast.

5.3 Models

We evaluated five GPT-2-vocabulary checkpoints:

Alias	Stream mode	Scale class	Notes
SS-71	single-stream	71M class	matched baseline
C-71	CASCADE	71M class	matched CASCADE pair
CP-71	CASCADE + PLS	71M class	same-scale variant
CP-80	CASCADE + PLS	80M+ class	12-layer model
CP-124	CASCADE + PLS	GPT-2-base class	12-layer model

The matched pair of greatest interest is:

- **SS-71**
- **C-71**

5.4 Sweep

- Layers: 0–5
- Scales: $\{-2, -1, -0.5, 0.5, 1, 2\}$
- Target and off-target evaluation: every source-case direction was tested against all three target cases

5.5 Metrics

Local gap shift. At the first divergent answer token, we measure

$$\Delta_{\text{local}} = (\text{logit}_A - \text{logit}_B)_{\text{intervened}} - (\text{logit}_A - \text{logit}_B)_{\text{baseline}} \,.$$

Total gap shift. For the full answer choices,

$$\Delta_{\text{total}} = (\text{score}(A) - \text{score}(B))_{\text{intervened}} - (\text{score}(A) - \text{score}(B))_{\text{baseline}} \,.$$

Tail fraction. Tail fraction summarizes how much of the total shift is not explained by the local answer-token shift under the current decomposition.

Off-target drift. For a source-case direction, we compute the same local shift on the other two cases and compare that drift to the strongest on-target shift.

6 Results

6.1 Baseline behavior

All five large models were baseline-correct on all three panel cases. This is already an improvement over the reduced-vocabulary small-model run, where one coreference case failed at baseline due to tokenization collapse.

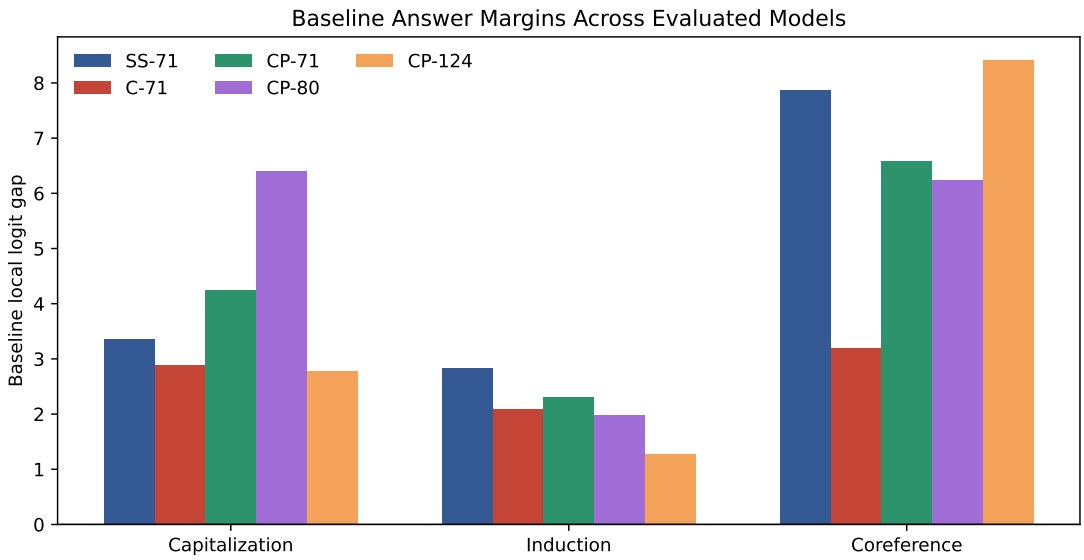

Figure 1: Baseline local answer margins for all evaluated models. All five large models are baseline-correct on the three-case panel before steering is applied.

6.2 Main steering result

Across all five models and all three cases:

- sign correctness was perfect in the tested sweep;
- locality was effectively perfect under the current metric;
- the direct vocabulary intervention remained bidirectional and causal.

This supports the narrow claim that direct vocabulary steering is not merely a small-model artifact.

6.3 Matched 71M comparison

The most informative comparison is the matched 71M pair:

- **SS-71** (single-stream)
- **C-71** (CASCADE)

Best absolute on-target local shifts:

Model	`caps_005`	`induction_009`	`coref_010`
single-stream 71M	2.48	1.15	0.94
CASCADE 71M	3.44	1.12	1.27

Interpretation:

- CASCADE is slightly stronger on capitalization and coreference.

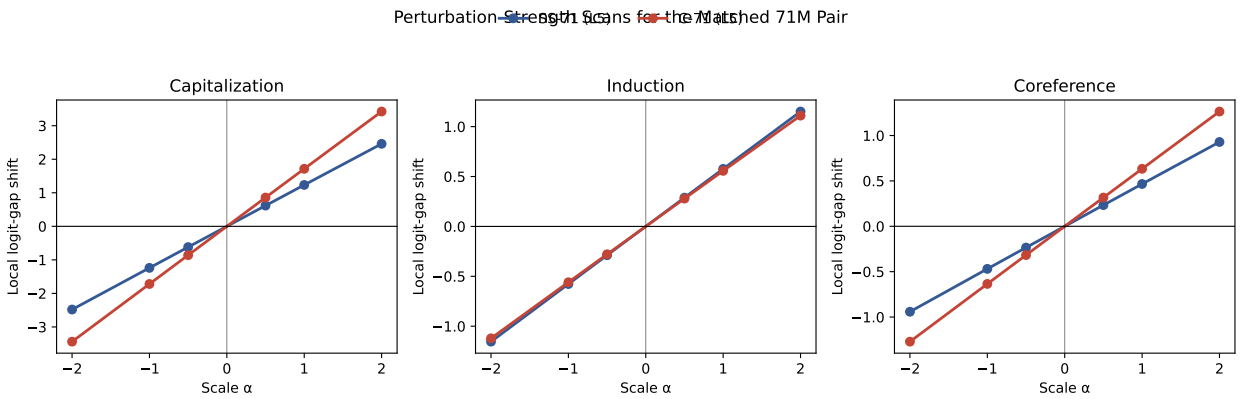

Figure 2: Perturbation-strength scans for the matched 71M pair. The curves show sign-consistent, bidirectional local steering across all three cases.

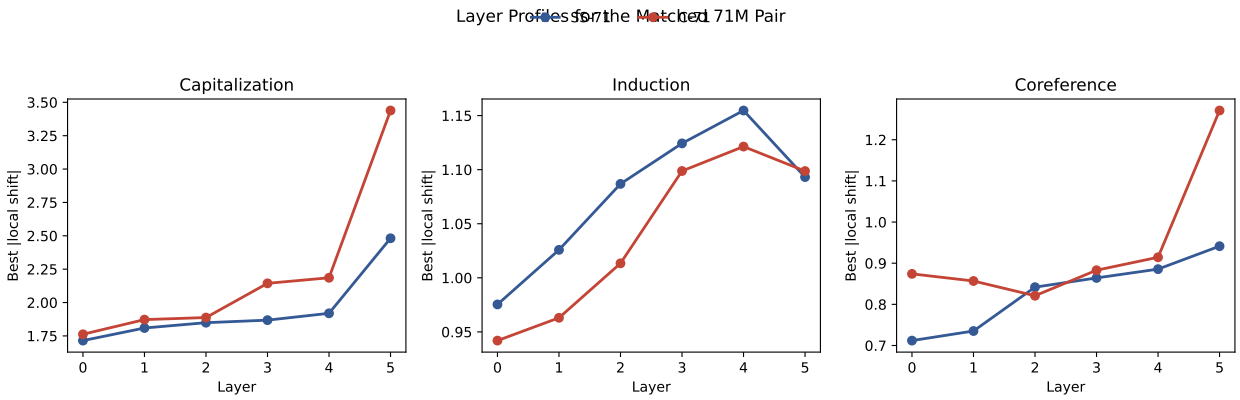

Figure 3: Best absolute local steering effect by layer for the matched 71M pair. The largest effects occur in later layers, with CASCADE modestly stronger on two of the three cases.

- Induction is essentially tied.
- The result favors CASCADE, but not strongly enough to support a large architecture-superiority claim.

6.4 Off-target drift

For the matched 71M pair, mean off-target drift is small relative to the strongest on-target effect:

Model	Source case	Best on-target	Mean off-target / best on-target
single-stream 71M	`caps_005`	2.48	0.033
single-stream 71M	`induction_009`	1.15	0.026
single-stream 71M	`coref_010`	0.94	0.072
CASCADE 71M	`caps_005`	3.44	0.022
CASCADE 71M	`induction_009`	1.12	0.072
CASCADE 71M	`coref_010`	1.27	0.098

This is good enough to say that the learned effect is not simply a global blunt perturbation.

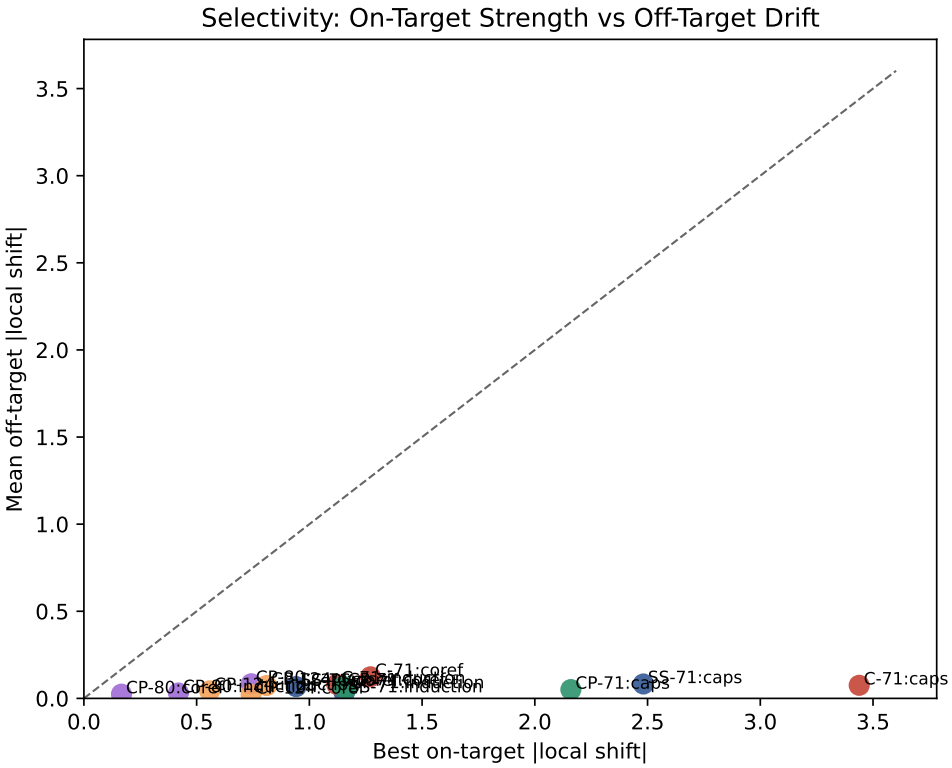

Figure 4: Selectivity summary across all evaluated models. Each point compares the strongest on-target effect for a source case with the mean off-target drift caused by that same direction. Points well below the diagonal indicate reasonably disciplined steering.

6.5 Larger 12-layer CASCADE models

The larger CASCADE+PLS models remained baseline-correct but showed smaller steering magnitudes at the tested scales than the 71M pair. This likely reflects intervention mismatch rather than disappearance of the effect. Plausible reasons include:

- the layer subset 0–5 no longer targets the most responsive depth range;
- the same scales are too conservative for larger representations;
- the chosen contextual injection site may be suboptimal for the 12-layer models.

7 Discussion

The most important point is that the direct-vocabulary effect survives contact with larger models. This is the main empirical advance over the earlier small checkpoint work.

The second point is methodological. Once the experiment is moved to raw GPT-2 tokenization and to stronger checkpoints, the intervention becomes much cleaner and off-target drift becomes substantially smaller. This means that the current bottleneck is no longer hardware or even model quality; it is benchmark panel quality.

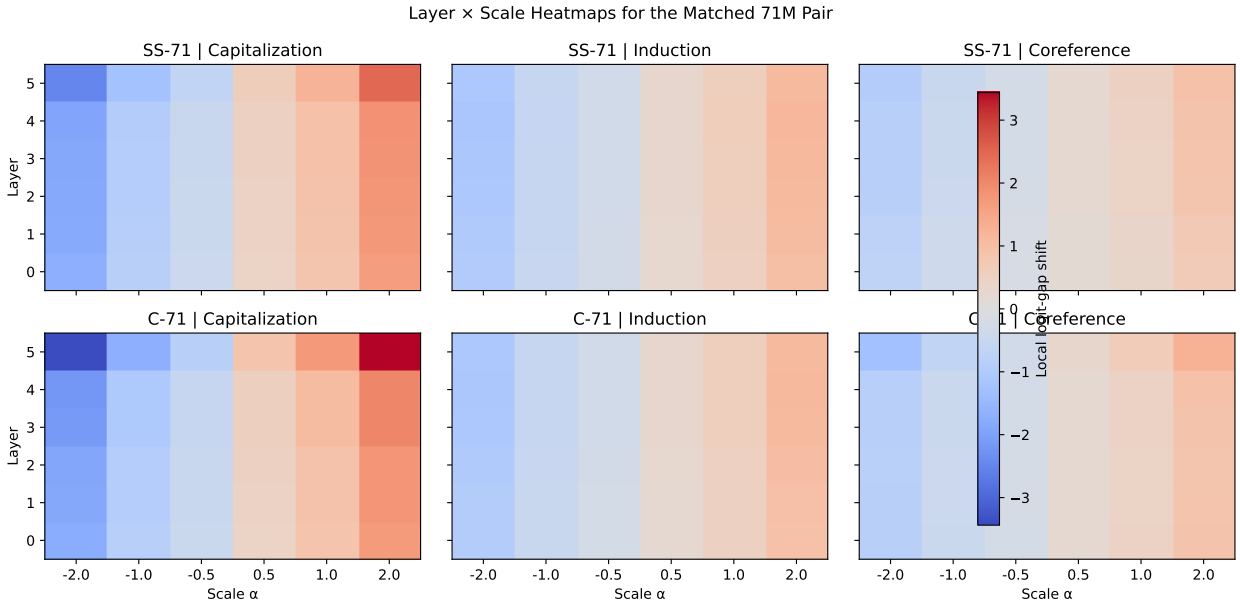

Figure 5: Layer × scale heatmaps for the matched 71M pair. These panels summarize sign symmetry, layer preference, and relative effect magnitude in a single view.

The third point is architectural. The matched 71M pair suggests that CASCADE is at least competitive and perhaps modestly cleaner than the single-stream model for direct vocabulary steering. However, the gap is not large. At this stage the best-supported claim is:

> Direct vocabulary steering works in both matched single-stream and CASCADE models, with a slight advantage for CASCADE on some tasks.

8 Limitations

1. **Tokenizer cleanliness is not fully solved.** `caps_005` remains a dirty capitalization case and should not anchor a publication-quality benchmark.
2. **The panel is too small.** Three cases are enough for feasibility, not enough for a robust claim about the generality of the effect.
3. **Layer specificity is still weak.** The strongest effects remain concentrated in late layers for the 71M pair rather than in a sharply localized intervention band.
4. **The larger 12-layer models need retuning.** The current sweep was held fixed to preserve comparability, not to optimize each model family.

9 Conclusion

The present evidence supports a positive but narrow conclusion: exact vocabulary-space contrasts can be used as causal steering directions in larger dual-stream and CASCADE-family models, and this effect remains local and reasonably selective under the current evaluation. The best

current evidence comes from the matched 71M GPT-2-vocabulary pair, where CASCADE is slightly stronger than the single-stream baseline but not decisively so.

The immediate next step is straightforward. Replace the dirty capitalization case with a tokenizer-clean contrast, expand the panel modestly, and rerun the matched 71M comparison. That is the shortest path to a memo that could support a cleaner literature-style steering claim.

10 Artifacts

Primary artifact directory:

- `runs/direct_vocab_large_models_cpu/`

  Key files:

- `runs/direct_vocab_large_models_cpu/direct_vocab_steering.json`
- `runs/direct_vocab_large_models_cpu/report.md`
- `scripts/run_direct_vocab_steering.py`